Zonglin Di

Contact Information	Tongji University, 1239 Siping Road Yangpu District, Shanghai, P. R. China, 201800	(+86) 185-1669-7168 1452640@tongji.edu.cn

Research Interest

Self-supervised learning, Weakly-supervised learning, Efficient AI, Computer Vision, Automatic control with graph neural network (GNN)

Education

University of California San Diego, Gilman Dr, La Jolla, CA, USA

Master of Science, Electrical and Computer Engineering, Sept 2020 to June 2022

- Supervisor: Prof. Xiaolong Wang

Tongji University, Shanghai, China

Bachelor of Engineering, Control Science and Engineering, Sept 2014 to July 2019

- GPA: 4.27/5.0 (87.7/100)
- Related Courses: Higher Mathematics, Linear Algebra, Possibility Theory & Statistics, Combinatorial Mathematics, Pattern Recognition, Object-Oriented Programming, Database Systems, Operating System, Compiler Principle, Digital Signal Processing, Digital Image Processing, Analog/Digital Circuits, Signal & Systems

Research Experience

Research Assistant of Xiaolong Wang's Lab June 2020 to Present

- Conduct self-supervised learning research on detection problem in long video scenario.
- Advisor: Prof. Xiaolong Wang & PhD. Xin Wang (Berkeley)

Research Assistant of Deep Learning Laboratory June 2018 to June 2020

- Perform large-scale roadmap extraction training on 40+ GPUs distributed training and rough labeling to study the limitation of weakly-supervised learning in semantic segmentation
- Improve the performance of road extraction using deep learning and GPS data, allowing 5% increase in accuracy and 40% boost in generalization ability for new area prediction
- Advisor: Prof. Yin Wang, PhD of University of Michigan

Research Intern in Intel Asia & Pacific R & D Oct 2017 to Mar 2018

- Improved the performance of existing semantic segmentation algorithms for removing human portrait from an image and patching the vacant area with variational auto encoders (VAE)
- Directed by Mr. Jiajie Yao

Research Assistant of State Lab of Embedded System & Service Computing, Ministry of Education Dec 2014 to Sept 2019

- Developed a novel method to facilitate the detection of Internet addiction disorder (IAD) so as to reduce the subjectivity compared to the traditional methods
- Completed a systematic test, showing the new model's capability to identify optimal feature combination in four questionnaires, with the accuracy of IAD labeling reaching 96.2%
- Advisor: Prof. Asoke K. Nandi, *IEEE Fellow*

Research Assistant of State Key Lab of Management & Control for Complex Systems, Ministry of Education Sept 2016 to Sept 2019

- Developed a hybrid method for binary and multi-class imbalance classification based on blending support vector machine (SVM) and clustering algorithms and using density peak as the metric
- Incorporated data distribution and data imbalance simultaneously for improving the imbalance classification of SVM

- Supervisors: Prof. Mengchu Zhou, *IEEE Fellow* & Prof. Qi Kang, *IEEE Senior Member*

Conference Publications

1. **Zonglin Di**, Qi Kang, Daogang Peng and Mengchu Zhou, "Density Peak based Pre-clustering Support Vector Machine for Multi-class Imbalanced Classification"
   IEEE Conference on System Man and Cybernetics (SMC), pp. 27-32, 2019
2. Tao Sun, **Zonglin Di**, Pengyu Che, Chun Liu and Yin Wang, "Leveraging Crowdsourced GPS Data for Road Extraction from Aerial Imagery"
   IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7509-7518, 2019
3. Tao Sun, **Zonglin Di** and Yin Wang, "Combining Satellite Imagery and GPS Data for Road Extraction"
   2nd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI), pp. 29-32, 2018
4. **Zonglin Di**, Siya Yao, Qi Kang and Mengchu Zhou, "AP-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification"
   IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 681-686, 2018
5. **Zonglin Di** and Qi Kang, "ADPclust-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification" (Abstract Accepted)
   IEEE International Conference on Networking, Sensing and Control (ICNSC), 2018
6. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, "Detection of Internet Addiction Disorder Based on Personality Questionnaires of Chinese College Students and SVMs"
   10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (ICSP-BMEI), pp. 1-6, 2017
7. Xiaoliang Gong, Bozhong Long, Kun Fang, **Zonglin Di**, Yichu Hou and Lei Cao, "A Prediction Based on Clustering and Personality Questionnaire Data for IGD Risk: A Preliminary Work"
   12th International Conference on Neural Computation, Fuzzy System and Knowledge Discovery (ICNC-FSKD), pp. 1699-1703, 2016

Journal Publications

1. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, "Internet Addiction Disorder Detection of Chinese College Students Using Several Personality Questionnaire Data and Support Vector Machine"
   Addictive Behavior Reports, Elsevier, 2019

Teaching Assistant

Python Programming (10064701) — Sept 2018 to Nov 2018
Deep Learning (10068501) — Mar 2019 to June 2019
Department of Computer Science and Technology, Tongji University
Supervised by Prof. Yin Wang

Open Source Hardware and Programming (55010501) — Sept 2018 to Jan 2019
School of Design and Innovation, Tongji University
Supervised by Prof. Xiaohua Sun

The Introduction to Deep Learning — March 2019 to June 2019
Department of Computer Science and Technology, Tongji University
Supervised by: Prof. Yin Wang

Intern Experiences

Full-Stack Software Intern in National Instruments, Shanghai April 2018 to April 2019

Established framework of Auto Build System in the Academic Group using React.js, Flask, and JQuery

Implemented frontend and backend within one-month timeframe independently

Directed by Mr. Dasheng Xu

Software Intern in National Instruments, Shanghai July 2017 to Oct 2017

Diadem data plug-in from German and Japanese customers using C++

Directed by Ms. Mavis Jin

Software Intern in Microsoft, Shanghai July 2016 to Sept 2016

Built ASP.NET WebSite and the backend engine

Directed by Mr. Tarry Wang and Alex Zhang

Technical Marketing Intern in Jikexueyuan, Beijing July 2015 to Sept 2015

Solved the Android issues in the community

Selected Awards

- The 2nd Price, 30th National "Challenge Cup" Academic Science and Technology Competition Nov 2019
- Special Award, 30th "Challenge Cup" Academic Science and Technology Competition Shanghai June 2019
- The Honorable Prize, 2019 Mathematical Contest in Modeling (MCM) Feb 2019
- The 2nd Price, Shanghai Collegiate Maker Competition Jan 2018
- The 3rd Prize, China Undergraduate Mathematical Contest in Modeling (Shanghai Division) Nov 2017
- Social Practice Scholarship, Tongji University Sept 2017
- The 3rd Prize, The 5th "Taidi Cup" Data Mining Contest Sept 2017
- The 3rd Prize, The 15th Digilent Design Contest July 2017
- The 3rd Prize, The 1st "Huichuangqingchun" Culture & Creativity Project Mar 2016

Selected Presentations

- SMC 2019, Bari, Italy Oct 2019
- CVPR 2019, Long Beach, USA June 2019
- GeoAI 2018, Seattle, USA Nov 2018
- SMC 2018, Miyazaki, Japan Oct 2018
- ICNSC 2018, Zhuhai, China Mar 2018
- ICSP-BMEI 2017, Shanghai, China Sept 2017
- ICNC-FSKD 2016, Changsha, China Aug 2016
- Excellent Undergraduate, Tongji University Sept 2015, 2016, 2017

Selected Projects

Heart Disease Detection using BNN and Digilent PYNQ

3-Layer Convolutional Neural Network implemented in FPGA

Accelerate the speed of training and prediction in a very large scale

Directed by Prof. Wei Zhang

National College Student Innovation Project

Record the action using frame-difference method and upload it to the DropBox in Python and OpenCV on Raspberry Pi

Directed by Prof. Maoheng Sun

5th "Taidi" Data Mining Contest

Nature Language Processing Project using CNN, SVM and KNN

CNN is implemented in TensorFlow and its accuracy is about 89%

Scrapy is used for language material scraping

Directed by Prof. Qi Kang

Shanghai College Student Innovation Project

Intelligent 3D Dressing System with Voice Control Based on Kinect and LabVIEW
Using ASP.NET API and LabVIEW to call the functions of Windows Cortana
A Chinese Patent is in progress Directed by Prof. Zhiming Zhang & Youling Yu

Microsoft-Tongji Innovation Training Project
Participate in the related course organized by Tongji University and Microsoft.
Develop a cross-platform online App using Xamarin and Twilio
Present the result of our group in Microsoft Cooperation, Zizhu, Minhang District on June 17th, 2016.
Directed by Mr. Han Xia

Student Innovation Training Project
Develop a OA System Using C#
Regular Expressions are used in this project
Directed by Ms. Zhihua Wei

Selected Social Experience

IEEE CVPR 2019 Student Volunteer June 15 to June 21, 2019
Helped guide the way to the different session for the participants
Helped organize the welcome party

Virtual Instruments Club Sept 2015 to Sept 2018
Member of Technical Department (Sept 2015 to Jan 2016) → **Minister of Technical Department** (Jan 2016 to Jan 2017) → **Vice President** (Jan 2017 to Sept 2017) → **President** (Sept 2017 to Sept 2018)
Collaborated with National Instruments (Great China Area) in organizing multiple on-campus events at Tongji University
Delivered 20+ presentations to 200+ students

Google Developer Group Dec 2016 to Sept 2018
Invited to Google Developer Days in 2016 and 2017
Organized the Google DevFes 2017
Organized the Google TensorFlow Technical Salon on the application of deep learning in computer vision and natural language processing

Unity Unite, Shanghai and Beijing May 2017 and May 2018
Keynote and Tech Conference Track Support

Reviewer Experience

- The 2nd International Conference on Mechanical, Electric and Industrial Engineering
- ACM SIGSPATIAL 2019 (External)